

Community-driven AI: Empowering people through responsible data-driven decision-making

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ABSTRACT

The increasing integration of AI and data-driven technologies in various sectors of society presents both opportunities and challenges, particularly regarding ethical, fair, and inclusive applications. This one-day workshop aims to convene researchers, practitioners, policymakers, and other community members to explore the complexities of community-driven AI with the aim to promote responsible and sustainable practices in the AI ecosystem. Drawing on interdisciplinary perspectives, participants will engage in discussions focused on themes such as understanding stakeholder relationships and positions in AI systems, evaluating community impact in AI ecosystems, examining policy and intervention implications, and devising strategies for effective partnership and collaboration with communities. The workshop will provide a platform for sharing experiences, exchanging ideas, and fostering long-lasting collaborations. By collectively addressing the challenges and opportunities in community-driven AI, we aspire to build an active community that supports ethical, inclusive, and sustainable AI-driven solutions, empowering communities in an increasingly data-driven world.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

Community-driven AI, AI Ecosystem, Responsible AI

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1 INTRODUCTION

The rapid advancement of Artificial Intelligence (AI) and data-driven technologies has created unprecedented opportunities for enhancing various aspects of society, including youth online safety [9, 15], child welfare system [3, 6, 10, 11], human trafficking detection [8, 14], health systems [1] and more. However, harnessing the full potential of AI and data for community empowerment requires a deep understanding of the ethical, social, and technical challenges that come with their application. For this reason, the CSCW community has hosted a series of workshops and panels focused on reflecting on the implications of designing technologies building on community-oriented methodologies (e.g., participatory design [16]), centering impacted communities when developing public interest technologies [13], and learning from grassroots politics of collaboration to build more equitable technologies [4].

In parallel, there has been an increased interest in developing community-oriented and participatory mechanisms in promoting awareness of algorithmic systems [5], learning about the concerns of communities affected by algorithmic decision-making systems [2], and contesting the outcomes of such systems [7]. However, most of these previous efforts tend to focus on community-oriented approaches that aim to intervene or inform specific aspects of designing, deploying, and evaluating AI systems rather than having a more holistic approach. Therefore, in this workshop, we propose to consider how different community members involved in the AI ecosystem might improve policy initiatives and systemic practices when developing AI-driven decision-making systems.

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Figure 1: Overview of AI stakeholders adapted from Stahl [12]

2 DEFINING THE AI ECOSYSTEM

We characterize the AI ecosystem by drawing from the categorization of *AI ethics stakeholders* definition proposed by Stahl [12]. Such categorization includes 1) policy-oriented bodies, 2) other organizations, and 3) individuals. While each of these stakeholders plays a critical role in developing, deploying, and using AI systems, Stahl also raises the concern that the most important stakeholders are those groups of people who are either not actively interested in shaping AI or that do not have a way to share their input but are still affected by the outcomes of AI systems [12]. By using Stahl’s overview of AI stakeholders, we aim to illustrate the complexity and often contradictory existence of various perspectives in the development of AI systems.

3 WORKSHOP THEMES AND OPEN QUESTIONS

Workshop attendees will work together to address several themes focused on understanding and tackling the challenges and opportunities of community-driven AI. We aim to develop a deeper understanding of the ethical, social, and technical implications of AI systems in community settings. To follow, we outline the five themes and pose open questions that will stimulate discussions and guide the workshop activities.

- (1) Understanding the varying relationships and positions of different community stakeholders (members, workers, advocates) within public sector AI systems. This theme focuses on the complex web of relationships among community stakeholders, including members, workers, and advocates, as they engage with AI systems in the public sector. By examining these relationships, we can identify potential power imbalances and ensure more equitable decision-making processes. Open questions:

- How do different community stakeholders (members, workers, advocates) interact with and influence AI systems?
 - What power dynamics exist among these stakeholders?
 - How can we address potential imbalances?
- (2) Analyzing the AI ecosystem from data labeling to design, development, evaluation, deployment, algorithm auditing, and maintenance. Understanding this ecosystem enables us to identify challenges and opportunities to expand community participation and improve transparency, accountability throughout the AI lifecycle. Open questions:
 - How do various stages of the AI lifecycle impact communities?
 - What challenges arise during data collection, model development, and system deployment?
 - How can we ensure community voices are heard throughout the process?
 - (3) Exploring policy and intervention implications of AI in community settings. Understanding the policy and intervention implications of AI-driven technologies in community contexts allows us to identify ethical considerations, potential harms, and strategies to address these challenges. Open questions:
 - How do AI-driven technologies affect communities in terms of policy and intervention?
 - What are the ethical considerations when implementing AI systems in community contexts?
 - How can we address potential harms?
 - (4) Effective strategies for partnering and collaborating with communities, with a focus on long-term sustainability. Establishing trust, understanding, and mutual benefit among stakeholders is essential for effective collaboration. Current approaches, such as co-design, participatory design, and

community-engaged research, aim to foster these positive relationships. However, challenges remain in ensuring clear communication, cultivating a sense of shared ownership, and addressing power imbalances between stakeholders. These aspects are crucial for building strong, long-lasting partnerships in community-driven AI research and development. Open questions:

- What approaches best support collaboration between AI researchers, practitioners, and communities?
 - How can we create sustainable partnerships that ensure lasting benefits for all stakeholders?
- (5) Developing theoretical frameworks for community-driven AI research. It provides a foundation for researchers and practitioners to build upon when designing, developing, and evaluating AI systems. It requires synthesizing different disciplinary perspectives, addressing the complex socio-technical nature, and creating frameworks that are flexible enough to accommodate diverse community contexts. Open questions:
- How can we establish robust, interdisciplinary frameworks that guide community-driven AI research?
 - What theoretical perspectives can inform the design, development, and evaluation of AI systems to ensure they are ethical, inclusive, and sustainable?

4 WORKSHOP GOALS

The aim of this workshop is to bring together a diverse group of researchers, practitioners, policymakers, and community stakeholders to explore the intersections of AI, data, and social spaces. By creating a space for interdisciplinary discussions, this workshop will foster the exchange of ideas, techniques, and best practices to address the challenges and opportunities that arise in community settings when AI and data-driven decision-making are employed.

5 CALL FOR PARTICIPATION

We invite researchers, practitioners, policymakers, community partners, and other stakeholders with an interest in community-driven AI, to join us in this interactive virtual workshop. This one-day synchronous online workshop aims to create a collaborative environment where participants can engage in interdisciplinary discussions, share their experiences, and contribute to the development of strategies and frameworks for harnessing the potential of AI in community settings.

Interested participants are asked to submit a brief statement of interest, demonstrating their alignment with the workshop goals. Submissions can take one of the following forms:

- A short bio with a statement of motivation/interest for attending the workshop.
- A 2-page position paper (includes figures; excludes references) discussing one or more of the workshop themes.
- A case study showcasing relevant work that contributes to the workshop themes.

We encourage submissions from individuals of diverse backgrounds, expertise, and experiences. Submissions should be in PDF format, following the ACM Primary Article Templates (single column). The submission should include author names (not anonymous submission). Each submission will undergo a peer review by

two organizers, with acceptance based on the quality, relevance to the workshop themes, and the potential for meaningful contributions to the workshop discussions and goals.

6 LOGISTICS AND EQUITY

We will recruit 30-40 participants who are researchers, practitioners, policymakers, and other stakeholders with an interest in community-driven AI, to join us in this interactive virtual workshop. To ensure a wide range of perspectives, we will recruit participants through social media, email lists, and relevant networks. Our website ¹ advertises the CFP, provides details about the workshop and how to submit, introduces the organizers, and provides resources to those interested in reading more about this domain. Prior to the workshop, we will add workshop submissions (bios or abstracts) from consenting participants, a schedule of events, and information related to accessibility accommodation. Following the workshop, we anticipate continuing to use the website for community building by hosting outcomes of the workshops, white papers, and special calls for participation in events related to the topic. This workshop will take place via Zoom video conferencing. To encourage more intimate and focused conversations, Zoom breakout rooms will be employed for smaller group discussions. Additionally, we will set up a Slack channel for participants to engage in our workshop and continue discussions beyond the event.

7 VIRTUAL WORKSHOP ACTIVITIES AND SCHEDULE

This one-day synchronous online workshop will feature a range of activities designed to promote collaboration, discussion, and the sharing of experiences among participants interested in community-driven AI. To facilitate participant engagement, develop ideas, and document conversations, workshop attendees will be able to contribute to an online collaborative whiteboard. The schedule will include the following sessions:

- **Welcome/Introduction (30 minutes):** Brief introductions from organizers and participants on workshop structure, goals and interest in Community-driven AI. We will also engage participants in an icebreaker activity to help ease participants into the space and set a collaborative mood to the event.
- **Community-driven AI Panel with Q&A (45 minutes):** We will hold a panel on defining what Community-driven AI means and what are best practices and challenges within it. We will recruit panelists who have experience working with communities around AI use and development.
- **coffee/networking break (15 minutes)**
- **Building Community Activity (60 minutes):** Participants will form groups based on research interests. They will introduce themselves to their breakout groups, present their research and its impact on communities, and discuss their engagement methods. They will also reflect on the challenges raised by the panel and whether they face similar issues.
- **Lunch break (60 minutes)**

¹Workshop website link:
<https://sites.google.com/nd.edu/community-ai-cscw-2023/home>

- **Imagining Solutions Activity (120 minutes):** Participants will have the opportunity to discuss each of the 5 workshop themes with their small groups. Through a structured exercise, they will brainstorm strategies that have worked based on their experiences and explore the possibilities for sustaining their research and its benefits for communities. They will also share key takeaways from their research, valuable insights for the larger group, and propose future actions to continue the conversation.
- **coffee/networking break (15 minutes)**
- **Report/Synthesize (30 minutes):** All participants will reconvene to share what they discussed in their breakout rooms, including their 3 key valuable insights, challenges and/or opportunities for the future, resources, and takeaways to allow for ongoing engagement after the workshop.
- **Next Steps (30 minutes):** Organizers will set-up a template roadmap in advance with ideas and options for future collaborations (e.g., special journal submission, workshop submission at a specific conference, monthly coffee meet-ups). We will spend this time discussing with participants how we should continue conversations and elaborating on the roadmap with concrete steps and dates. Participants will be encouraged to lead efforts and/or participate in the activities to carry out the roadmap.

8 POST-WORKSHOP PLANS

After the virtual workshop we plan to have an in-person meet-up (for participants who attend CSCW in person) and publish a workshop summary for the community to continue conversations. The goal of the in-person CSCW conference meet-up is to facilitate networking, collaboration, and deeper discussions. The in-person meet-up will be held in a designated meeting space at the conference venue and will adhere to all necessary health and safety guidelines in place at the time of the event. Further, the organizers will collaboratively author a summary of the discussions that happened during the workshop. This will be published either as a blog post on Medium.com or an Interactions magazine submission. Furthermore, we will maintain our Slack channel as a communication platform for participants to continue discussions, share resources, and collaborate on projects related to the workshop themes. This will provide a convenient and accessible space for the community to stay connected and engaged beyond the workshop itself.

9 WORKSHOP ORGANIZERS

Our team of organizers is comprised of students, faculty, and industry professionals that bring unique expertise and vision to the topic of community-driven AI.

Ruyuan Wan (Main contact) is a PhD student at the University of Notre Dame in the Computer Science and Engineering Department. She works at the intersection of Human-Computer Interaction and Natural Language Processing with a social computing orientation. Her research focuses on how to promote online safety for marginalized populations.

Adriana Alvarado Garcia is a research scientist at IBM Research, working with the Responsible and Inclusive Technologies research team. She contributes to creating frameworks, methods, and tools

that help researchers and practitioners become more aware of the social, historical, and political contexts surrounding the technologies they design and deploy.

Devansh Saxena is a Presidential Postdoctoral Fellow at Carnegie Mellon University in the Human-Computer Interaction Institute. He studies algorithmic decision-making in the public sector, especially the child welfare system. His research examines collaborative governance labor where decisions are mediated by policies, practice, and algorithms.

Catalina Vajiac is a Ph.D. student at Carnegie Mellon University in the Computer Science Department. Her research is focused on AI and visualization for social good, including finding human sex trafficking online and better prioritization of financial resources to those at risk of falling into homelessness.

Anna Kawakami is a PhD student in the Human-Computer Interaction Institute at Carnegie Mellon University. She studies the design, use, and downstream impacts of AI systems on communities, workplace dynamics, and governance practices in real-world organizations (e.g., public sector agencies).

Logan Stapleton is a Ph.D. student in the GroupLens Lab at University of Minnesota and a visiting student at Carnegie Mellon University. His work focuses on harm, discrimination, and carcerality in algorithms used in social work domains, particularly child welfare and suicide prevention. He explores how to design to better support suicidal people online.

Haiyi Zhu is the Daniel P. Siewiorek Associate Professor of Human-Computer Interaction and the Director of HCI Undergraduate Programs at Carnegie Mellon University. She is interested in incorporating stakeholder values and social considerations into the creation and evaluation process of AI technologies. She has conducted several projects that focus on studying the design and social impacts of AI in child welfare, online content moderation, and mental health.

Kenneth Holstein is an Assistant Professor of Human-Computer Interaction at Carnegie Mellon University, where he directs the CoALA Lab. His work explores human-AI complementarity and AI-augmented work in settings such as social work, education, and design. He is interested in improving the way human-AI decision systems are designed, developed, and evaluated in real-world settings.

Heloisa Candello is a research scientist and a manager of the Human-centred & responsible tech, at the IBM Research laboratory in Brazil. Her main research interest is applying design methods to understand how people interact with Artificial Intelligence-based systems, particularly conversational systems. Heloisa leads and conducts research activities in HCI to understand the social impact, and motivation of using conversational technologies.

Karla Badillo-Urquiola is a Clare Boothe Luce Assistant Professor of Computer Science and Engineering at the University of Notre Dame. She leverages her interdisciplinary expertise to investigate technology-driven solutions that empower people and protect the well-being of youth in marginalized communities, drawing on participatory and critical design methods.

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